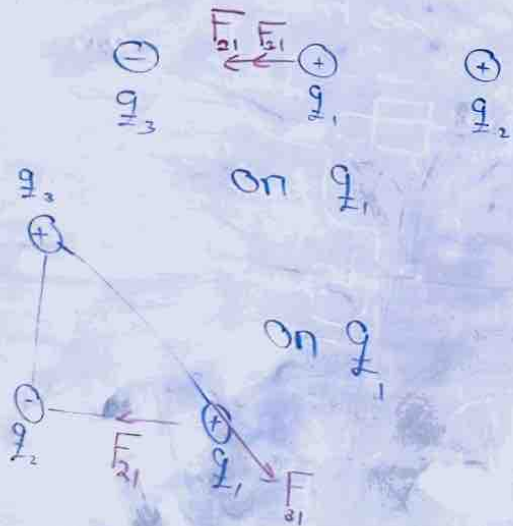
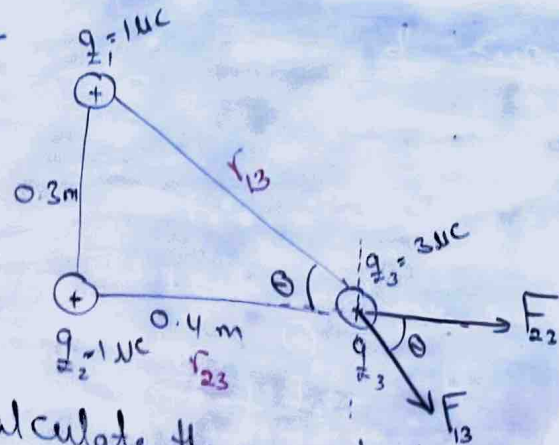


Principle of Superposition

This principle is used to calculate the total electric force on a give charge q_0 due to other charges $q_1, q_2, \dots$





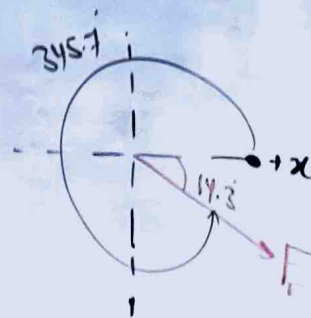
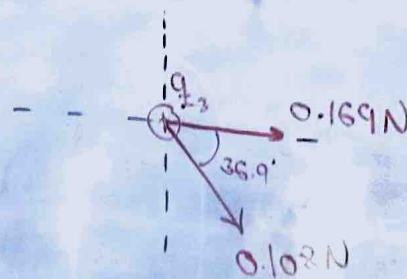
Calculate the magnitude and the direction of the resultant electric force on charge $q_3 = 3 \mu\text{C}$ due to $q_1 = q_2 = 1 \mu\text{C}$

$$\theta = \tan^{-1}\left(\frac{0.3}{0.4}\right) = 36.9^\circ$$

$$r_{13} = \sqrt{0.4^2 + 0.3^2} = 0.5 \text{ m}$$

$$F_{13} = k_e \frac{q_1 q_3}{r_{13}^2} = 9 \times 10^9 \times \frac{1 \times 10^{-6} \times 3 \times 10^{-6}}{0.5^2} = 0.108 \text{ N}$$

$$F_{23} = k_e \frac{q_2 q_3}{r_{23}^2} = 9 \times 10^9 \times \frac{1 \times 10^{-6} \times 3 \times 10^{-6}}{0.4^2} = 0.169 \text{ N}$$



$$F_x = 0.169 + 0.108 \cos 36.9^\circ = 0.255 \text{ N}$$

$$F_y = -0.108 \sin 36.9^\circ = -0.065 \text{ N}$$

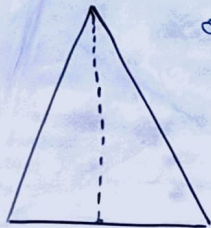
$$F = \sqrt{0.255^2 + (-0.065)^2} = 0.263 \text{ N}$$

$$\theta = \tan^{-1}\left(\frac{-0.065}{0.255}\right) = -14.3^\circ \text{ (forth)}$$

$$\theta_{rx} = 360 - 14.3 = 345.7^\circ$$

بسم الله الرحمن الرحيم
Physics II

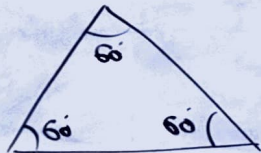
$$\begin{cases} mC = 10^{-3} C \\ \mu C = 10^{-6} C \\ nC = 10^{-9} C \\ pC = 10^{-12} C \end{cases}$$



$$\begin{array}{c} 0.5 \quad 1.8 \quad 2 \\ \leftarrow \quad \quad \rightarrow \end{array}$$

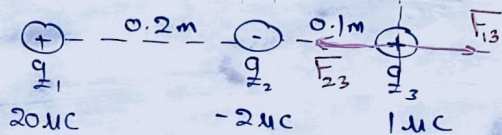
$$F_r = 2 - 1.8 - 0.5 = -0.3 N$$

$$[0.3 N, \theta = 180^\circ]$$



equilateral triangle
{ثلاثي متساوي الساقين}

Q4

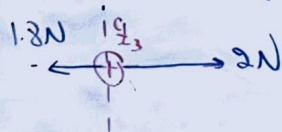


Find the magnitude and direction from $+q$ of the electric force on $q_3 = 1 \mu C$

$$F_{13} = K_e \frac{q_1 q_3}{r_{13}^2} = 9 \times 10^9 \times \frac{20 \times 10^{-6} \times 1 \times 10^{-6}}{0.3^2} = 2 N$$

0.2 + 0.1

$$F_{23} = K_e \frac{q_2 q_3}{r_{23}^2} = 9 \times 10^9 \times \frac{2 \times 10^{-6} \times 1 \times 10^{-6}}{0.1^2} = 1.8 N$$

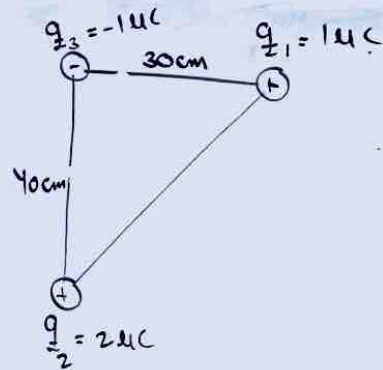
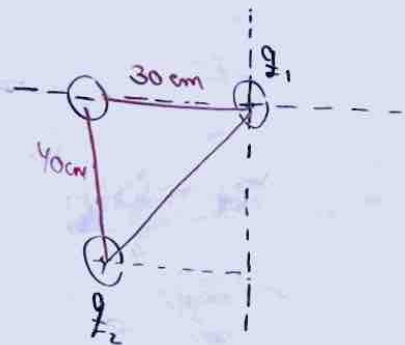


$$F_r = 2 - 1.8 = 0.2 N$$

$$[0.2 N, \theta = 0^\circ]$$

Physics II Q₁

Find the resultant electric force on $q_1 = 1 \mu\text{C}$ placed at origin due to two charges, $q_2 = 2 \mu\text{C}$ is placed at $(-30 \text{ cm}, -40 \text{ cm})$ and $q_3 = -1 \mu\text{C}$ is placed at $(-30 \text{ cm}, 0)$

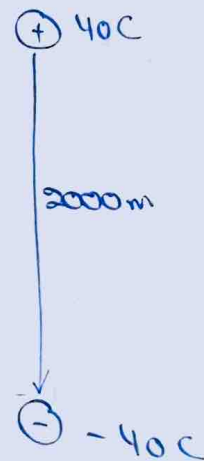
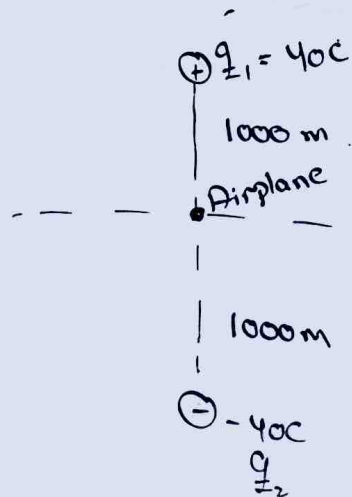
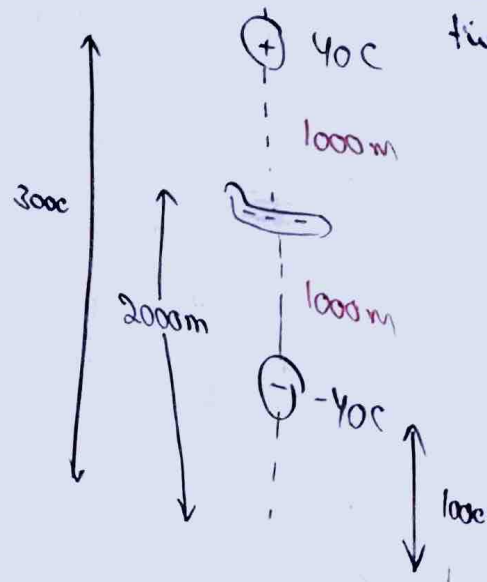


Q₂ Find the electric force on $q_2 = -3 \mu\text{C}$ placed at middle between two charges, $q_1 = 2 \mu\text{C}$ is placed at $(0, 60 \text{ cm})$ and $q_3 = 1 \mu\text{C}$ is placed at $(0, -20 \text{ cm})$

An airplane is flying at 2000m

if a charge concentration +40C is above the airplane at 3000m and -40C at 1000m

Find the electric force between the two charges.



$$F = k_e \frac{q_1 q_2}{r_{12}^2} = 9 \times 10^9 \times \frac{40 \times 40}{2000^2} = 36 \times 10^5 \text{ N}$$